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Subject: Workflow Automation

Assignment: BPMN Project Assignment

Assignment 1: Student Project Approval and Allocation

What the process is

- A student team sends a project proposal. It is checked, reviewed by the committee, and then a faculty guide is given to the team.
- It starts when the team decides to submit. It ends when a guide is confirmed, or the proposal is rejected or lapsed.

Lanes used

- Student / Team, Project Coordinator, Review Committee (with HoD), Project Management System.
- Faculty Guide is a separate pool because the guide is outside the process. We talk to the guide with messages.

Happy path

- Team submits proposal.
- System checks the form and team rules.
- Coordinator screens the proposal.
- System runs similarity check against old projects.
- Committee reviews and approves.
- System checks which guide is free (workload and domain).
- System reserves the guide and sends the request.
- Guide accepts.
- System saves the allocation and tells the student.
- End: project allocated.

Task types and why

- User task: where a person does the work (submit proposal, screen, committee review, coordinator decisions).
- Service task: where the system does it alone (validate form, similarity check, reserve guide, record allocation).
- Business rule task: for "which guide is free", because it is decided by rules (domain match, max load).
- Send task: when we send something out (request to guide, rejection reasons, notify student).

- Message catch event: to wait for the guide reply.

Gateways used and why

- Exclusive gateway (X): used when only one path can be taken. Examples: form valid or not, committee approve / revise / reject, guide found or not.
- Counter gateways like "Resubmissions < 2?" and "Attempts < 3?": these stop loops from going on forever.
- Event-based gateway: after sending the request to the guide. Whichever comes first, the guide reply or 3 days timer, decides the path.
- I did not need parallel or inclusive gateway in this one because the steps happen one after the other.

Sub-processes

- Guide allocation: all the guide finding steps are put in one box so the main diagram stays clean. A withdrawal event is attached to this box.
- Record allocation and notify: saving and notifying. An error event is attached so retry can be done.

Failure path register (Assignment 1)

F1 - Form not filled properly. Problem: Missing fields, wrong roll numbers. What I did: Form goes back to the team with the errors. They can resubmit only 2 times. After that the coordinator closes the case.

F2 - Team rule broken. Problem: Too many members, or student already in another team. What I did: Team is told and the process ends as "Invalid team". They have to make the team again.

F3 - Deadline missed. Problem: Team did not submit in time. What I did: Timer on the submit task. Team asks for late submission, coordinator says yes or no.

F4 - Same topic as old project. Problem: Similarity check score is too high. What I did: Coordinator checks the match. Either topic is changed and checked again, or it is rejected, or it is a false match and goes ahead.

F5 - Committee wants changes. Problem: Scope too big, objective not clear. What I did: Student revises and sends again. Only 2 rounds are allowed. After that the committee must approve or reject.

F6 - Student does not revise. Problem: No reply for 7 days. What I did: Timer of 7 days. Process ends as "Lapsed".

F7 - Committee rejects. Problem: Not possible to do, out of syllabus. What I did: Reasons are sent. Team gets one chance to give a new topic. Second time it ends as "Rejected".

F8 - Committee is late. Problem: No decision in time. What I did: After 7 days a reminder is sent. After 14 days it goes to the HoD and he decides.

F9 - No guide free. Problem: Domain does not match or guides are full. What I did: Coordinator gives guide by hand, or adds co-guide or outside guide, or puts team on waiting list (14 days, then check again).

F10 - Guide says no or does not reply. Problem: Busy, conflict, or no answer in 3 days. What I did: That guide is removed and the rule runs again. After 3 tries the coordinator gives the guide by hand.

F11 - Team or guide leaves after approval. Problem: Team breaks up, or guide goes on leave. What I did: Message event on the allocation part. The guide slot is released (compensation). If guide left, allocation starts again. If team left, it ends as "Withdrawn".

F12 - Notification or system fails. Problem: Email down or database error. What I did: Error event. System retries 3 times. After that the coordinator informs the team by hand.

Assignment 2: Logistics and Shipment Exception Management

What the process is

- A shipper books a parcel. It is picked up, moved through hubs and delivered to the recipient.
- It starts when the shipper books. It ends when parcel is delivered and closed, returned to sender, or closed as lost or cancelled.

Lanes used

- Shipper / Customer, Pickup Agent, Control Tower / Dispatcher, Hub / Warehouse, Delivery Agent, Recipient, Claims Department.

Happy path (normal flow)

- Shipper creates booking.
- System checks address, weight, service level and calculates price.
- Payment is taken and pickup is scheduled.
- Pickup agent collects the parcel and scans it.
- Two things happen together: tracking update to shipper, and the parcel moves to the hub.
- Both finish, then parcel is received and scanned at the hub.
- Checks are done if needed (customs, damage check), then parcel is sorted.
- Recipient is told parcel is out for delivery.
- Delivery agent delivers and takes proof of delivery.
- System closes the shipment and makes the invoice. End.

Task types and why

- User task: when a person does it (create booking, collect parcel, scan at hub, deliver).
- Service task: when the system does it (take payment, schedule pickup, close shipment).
- Business rule task: for booking check and price, because it follows rules.
- Send task: for notifications to shipper and recipient.

Gateways used and why

- Parallel gateway: after pickup, tracking update and transport happen at the same time. Then a second parallel gateway waits for both.
- Inclusive gateway: at the hub. A parcel may need customs, or damage check, or both, or none. So more than one path can be taken. Sorting always happens (default path).
- Exclusive gateway: for one-path choices like booking valid or not, damage found or not, insured or not, delivery outcome.
- Event-based gateway: used 2 times. In customs (wait for documents or 5 days timer) and when recipient is not home (wait for reply or 1 day timer).

- Counter gateways: "Corrections < 3?", "Reschedules < 2?", "Attempts < 3?" to limit loops.

Sub-processes

- Transport to destination hub: runs once for each hub leg (multi-instance). Delay, lost parcel and cancellation events are attached to it.
- Customs clearance: document check, request documents, wait.
- Claims handling: assess claim, then refund / replace / reship or reject claim.

Failure path register (Assignment 2)

E1 - Booking is wrong. Problem: Bad address, weight or service level. What I did: Goes back to shipper to correct. Max 3 times, then booking is rejected.

E2 - Pickup missed. Problem: Shipper not there or parcel not ready. What I did: Timer on pickup task. Pickup is booked again (max 2 times). After that booking is cancelled and fee is charged.

E3 - Scanner or system failure at hub. Problem: Scanner offline, data mismatch. What I did: Error event on the scan task. Supervisor enters the scan by hand, then records are matched again.

E4 - Delay on the way. Problem: Weather, vehicle breakdown, hub full. What I did: Delay message event. Customer is told and ETA is updated. Another vehicle or route is given. Transport keeps going.

E5 - Parcel lost. Problem: No scan for 48 hours. What I did: Timer starts a trace. If not found in 5 days it is marked "Lost" and compensation is paid. If found, it goes on from the join.

E6 - Shipper cancels after dispatch. Problem: Cancelled when parcel is moving. What I did: Message event stops the transport. Parcel is brought back and payment is refunded minus fee (compensation).

E7 - Customs problem. Problem: Documents missing or wrong. What I did: Documents are asked from the shipper. Wait for documents or 5 days timer. If not given, parcel is returned to sender.

E8 - Parcel damaged. Problem: Found at hub check. What I did: If insured, goes to claims (refund, replace or reship, or claim rejected). If not insured, damage is noted, shipper is told and delivery goes on.

E9 - Wrong address at door. Problem: Address not found or not complete. What I did: Error event on delivery task. Agent calls recipient and confirms address (extra charge if changed), then tries delivery again.

E10 - Recipient not there. Problem: No one at address. What I did: Up to 3 attempts. Between attempts wait for recipient reply or 1 day. After 3 attempts parcel is kept at depot for 7 days, then returned.

E11 - Refused or cash on delivery fails. Problem: Recipient refuses parcel or does not pay. What I did: Return to sender is arranged and charges are adjusted.